Architecture and Design

Obelisk

Prepared by: Filliciano Plong

Date: October 23, 2026

Version: 1.1

Readers: Filliciano Plong, sole developer. Helps decide module boundaries, the save path, and the export pipeline before code starts.

Contents

Revision history....2
1. Context and scope....3
2. Goals and constraints....3
3. Quality-attribute scenarios....4
4. Architectural style....4
5. Architecture views....6
6. Behavior and contract....13
7. Architecture decisions....22
8. Risks....26
9. Technology stack....26
10. Traceability....28
11. Open decisions....31
12. Maintenance and references....32

Revision history

Version	Date	Changes
1.0	October 23, 2026	First release of the architecture.
1.1	October 23, 2026	Applied the design, data model, and security review fixes. Based on SRS 1.1.

1. Context and scope

This document shows how Obelisk will meet its SRS. It cites requirement IDs from the SRS (Plong_Obelisk_SRS.docx, version 1.1) instead of repeating them.

Everything here is a proposal until accepted. Technology versions are unverified until the spikes in section 9 confirm them.

1.1 Actors

Actor	Role
User	Writes, saves, prints, and exports documents; one person in the student or professional role.
Windows	File Explorer, Open and Save dialogs, print spooler, clipboard, spell checker, DPAPI.
Microsoft Store	Installs, signs, and updates the MSIX package.

1.2 External dependencies

- Electron (Chromium and Node.js), React, TypeScript.
- A ProseMirror-based editor core (proposal, ADR-004).
- KaTeX for formulas; mathml2omml for Word equations (Should).
- Mermaid for the diagram block (Should, OD-05).
- Bundled fonts: the LibreOffice Latin set plus Dancing Script (OD-08).

1.3 Included and excluded

Included: SRS sections 6 and 7: every FR and NFR, Must and Should.

Excluded: The SRS out-of-scope list (section 2.2) and future features (section 13).

2. Goals and constraints

2.1 Top quality goals

Rank	Goal	Requirements
1	No lost or half-written files	NFR-10, NFR-11, NFR-12, NFR-13, NFR-14
2	Fast typing and opening in long documents	NFR-01, NFR-02, NFR-03, NFR-04
3	Safe with untrusted files and pastes	NFR-06, NFR-07, FR-09, FR-36
4	Faithful print and export	FR-62, FR-63, FR-64, FR-56

2.2 Constraints

Constraint	Source
Electron, React, and TypeScript	Developer decision
Microsoft Store MSIX is the only release channel	Developer decision; ADR-008
Solo developer; Shoulds cut in the SRS drop order	Project plan
English interface; Letter paper	SRS section 5

Constraint	Source
mathml2omml is LGPL and ships as a separate module	NFR-29

3. Quality-attribute scenarios

Each scenario names a stimulus and environment, the expected response, a measure, and how it is checked.

ID	Stimulus and environment	Response and measure	Verification	IDs	Component
QS-01	User types in a 50-page document with tables and formulas, on the reference machine.	Character shows within 50 ms; formulas re-render only when their source changes.	Test: input-latency trace	NFR-01, NFR-02	EditorCore, FormulaRenderer
QS-02	The app is ended or power is lost during Save or autosave of a large file.	The target holds the old bytes or the complete new file; 20 of 20 runs.	Test: kill during save	NFR-10, NFR-12, NFR-14	FileService
QS-03	User opens a crafted .obk file, or pastes HTML with scripts and remote images.	Nothing runs and no request is sent; a bad file opens nothing and stays unchanged.	Test: hostile sample set (section 6.10) under Node and in the app	FR-09, FR-36, NFR-06, NFR-07	FormatCodec, ClipboardBridge
QS-04	Another program changes an open file, then autosave fires.	Autosave is skipped; the user chooses Overwrite, Save As, or Cancel; nothing is overwritten silently.	Test: change the file in Notepad	FR-13, FR-15	AutosaveService, FileService
QS-05	User exports a document with 20 formulas to Word.	Each formula is an editable equation, or an image if conversion fails; done within 10 s.	Test: open in Word	FR-56, FR-57, NFR-04	ExportService
QS-06	Another Windows user on the same PC reads Obelisk's app data.	Saved signatures are unreadable without the owner's DPAPI key. Other apps of the same user are out of scope.	Inspection: read the file as another user	NFR-08, NFR-09	SignatureStore
QS-07	A script runs in a renderer (an assumed bug) and calls the IPC bridge with made-up handles, IDs, and documents.	Main refuses what it did not issue; nothing outside user-picked files and app data is read, written, or deleted.	Test: fake renderer under Node (H-30)	NFR-07, NFR-22	IPC bridge, FileService

4. Architectural style

Local-first modular monolith. One Electron main process owns all file, print, and export work. One sandboxed renderer per window owns the editor. They talk only over a narrow, typed IPC bridge

exposed by the preload script. Every disk write goes through one tested helper (ADR-001, ADR-002). Untrusted content stays away from Node, and no server is needed, which matches NFR-05.

Component	Process	Responsibility
Shell and Tabs	Renderer	Window regions, menus, tab strip, status bar (including Autosave paused), Preferences and About tabs, prompts.
EditorCore	Renderer	Document model, typing, shortcuts, slash menu, undo, selection, block moves, find and replace, page layout. New block IDs on paste and duplicate.
BlockViews	Renderer	Views for tables (with the TableMath module), images, columns, drawing, signatures, date chips, and diagrams (Mermaid, strict). Makes first-frame PNGs of animated images for export.
FormulaRenderer	Renderer	KaTeX rendering with fixed settings (section 6.4), live preview, errors; MathML and PNG for export.
ClipboardBridge	Renderer	Copy and paste of blocks. Pasted HTML is parsed inert and mapped into the schema; pasted Obelisk blocks are checked like an opened file.
AutosaveService	Renderer	Per-tab 30 s timer after the last edit. Saved tabs: calls file:save like Ctrl+S. Never-saved tabs: sends recovery:put. Pauses on conflict.
IPC bridge	Preload	The only path from renderer to main: named, typed calls. Main checks the sender and every argument; handles, never paths.
FileService	Main	Open, Save, and Export dialogs; the handle-to-path map; disk-change check; file activation and session restore; the shared atomic-write helper writeAtomic.
FormatCodec	Main	Encode and decode .obk files: archive caps, entry allowlist, schema, limits, image headers, and version conversion. Runs under Node.
ExportService	Main	Print, PDF, Word, and Markdown exporters behind one interface, export(format, document, assets, target); writes through writeAtomic.
SignatureStore	Main	Saved signatures: one PNG per file, encrypted with the async safeStorage API; writes through writeAtomic.
PreferencesStore	Main	Preferences and the session in app data; bad values fall back to defaults; writes through writeAtomic.
RecoveryService	Main	Recovery copies of never-saved tabs in app data, named by UUIDs that main makes; checked by FormatCodec on restore.

5. Architecture views

Each view is a PlantUML source in docs/diagrams/. Component names match the table in section 4.

5.1 Use case view

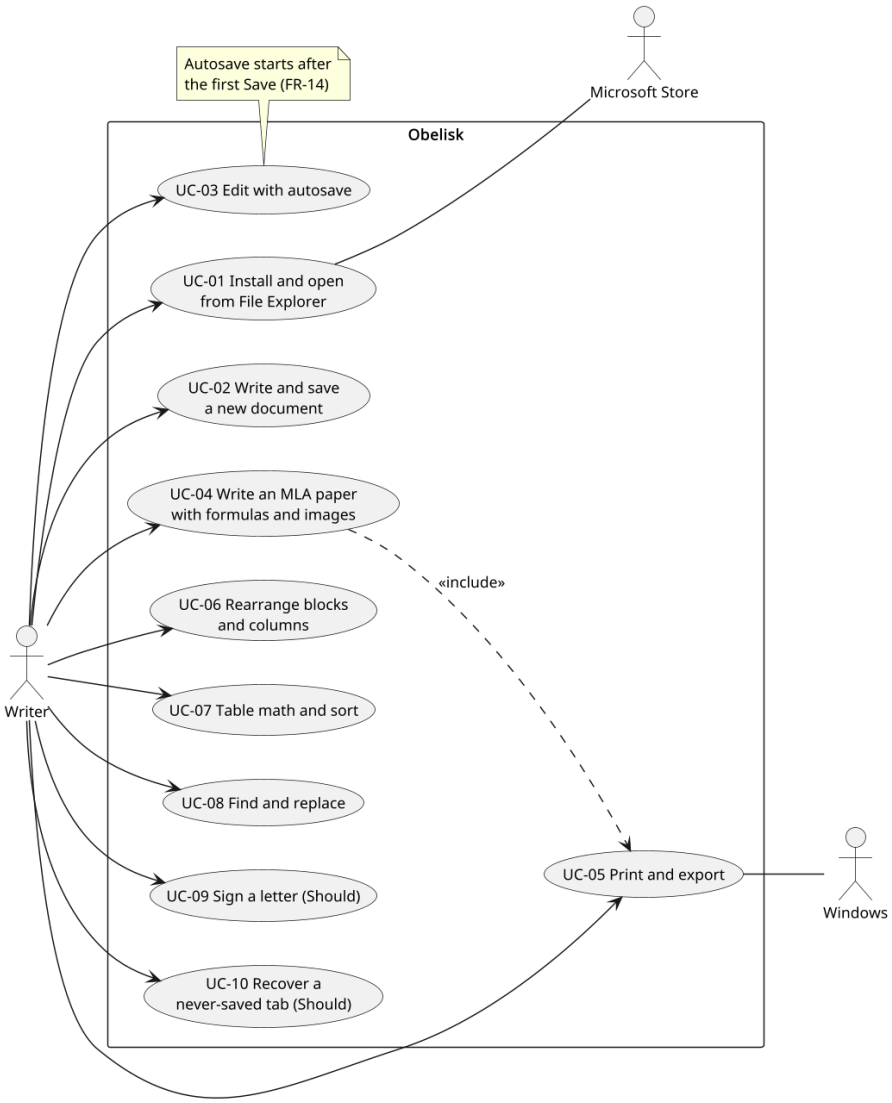


Figure 1. *Obelisk use cases*

Purpose: Shows who uses Obelisk for what; details are in SRS section 8.

Assumption: One user; Windows and the Store are supporting actors.

Limitation: No flows or errors; see the sequences.

5.2 Component view

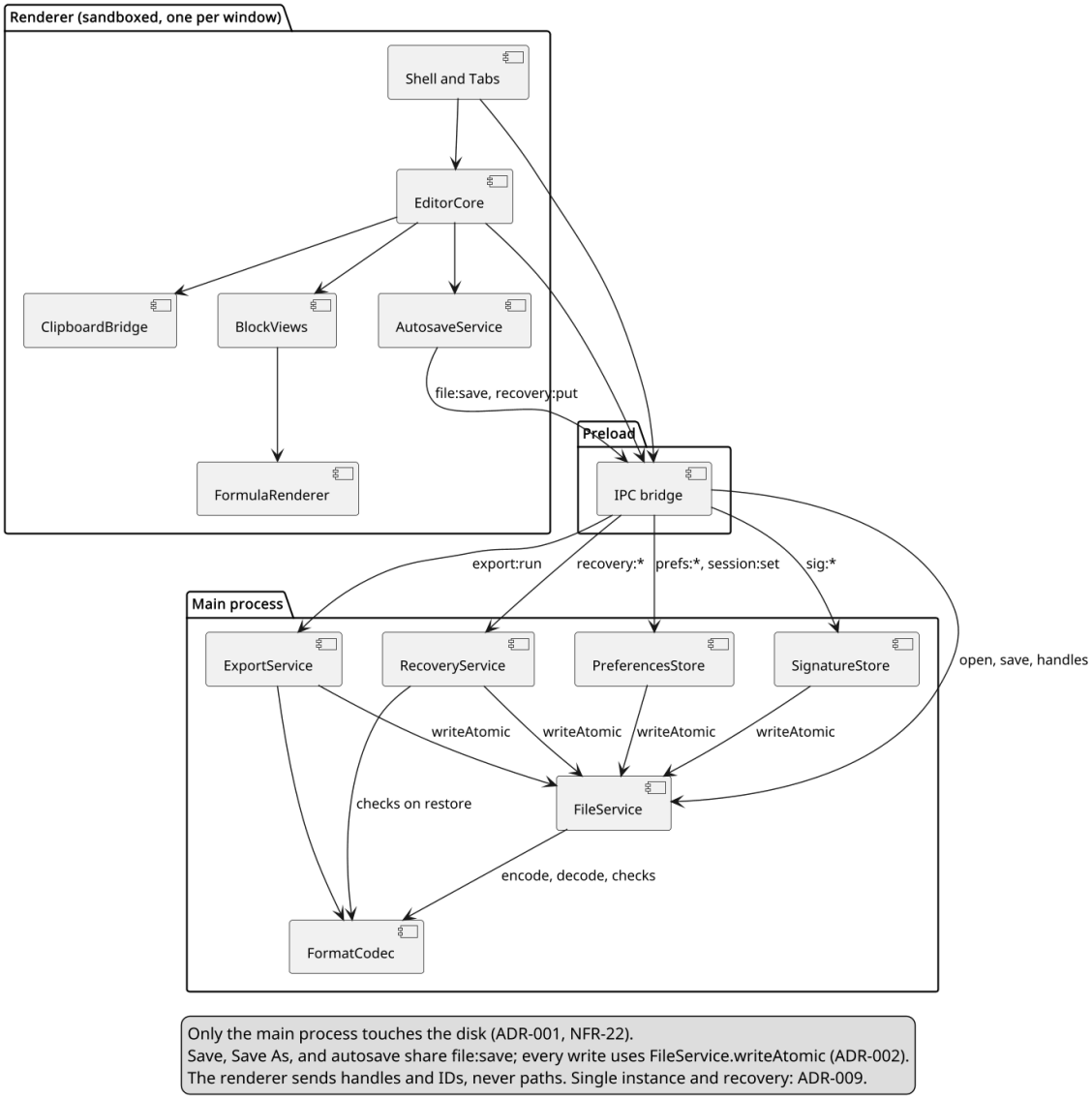


Figure 2. *Components and dependencies*

Purpose: Shows which component owns each responsibility, the one path to disk, and who uses writeAtomic.

Assumption: One window with many tabs; one renderer per window.

Limitation: Sub-modules such as TableMath and the exporters are not drawn.

5.3 Sequence: save and autosave

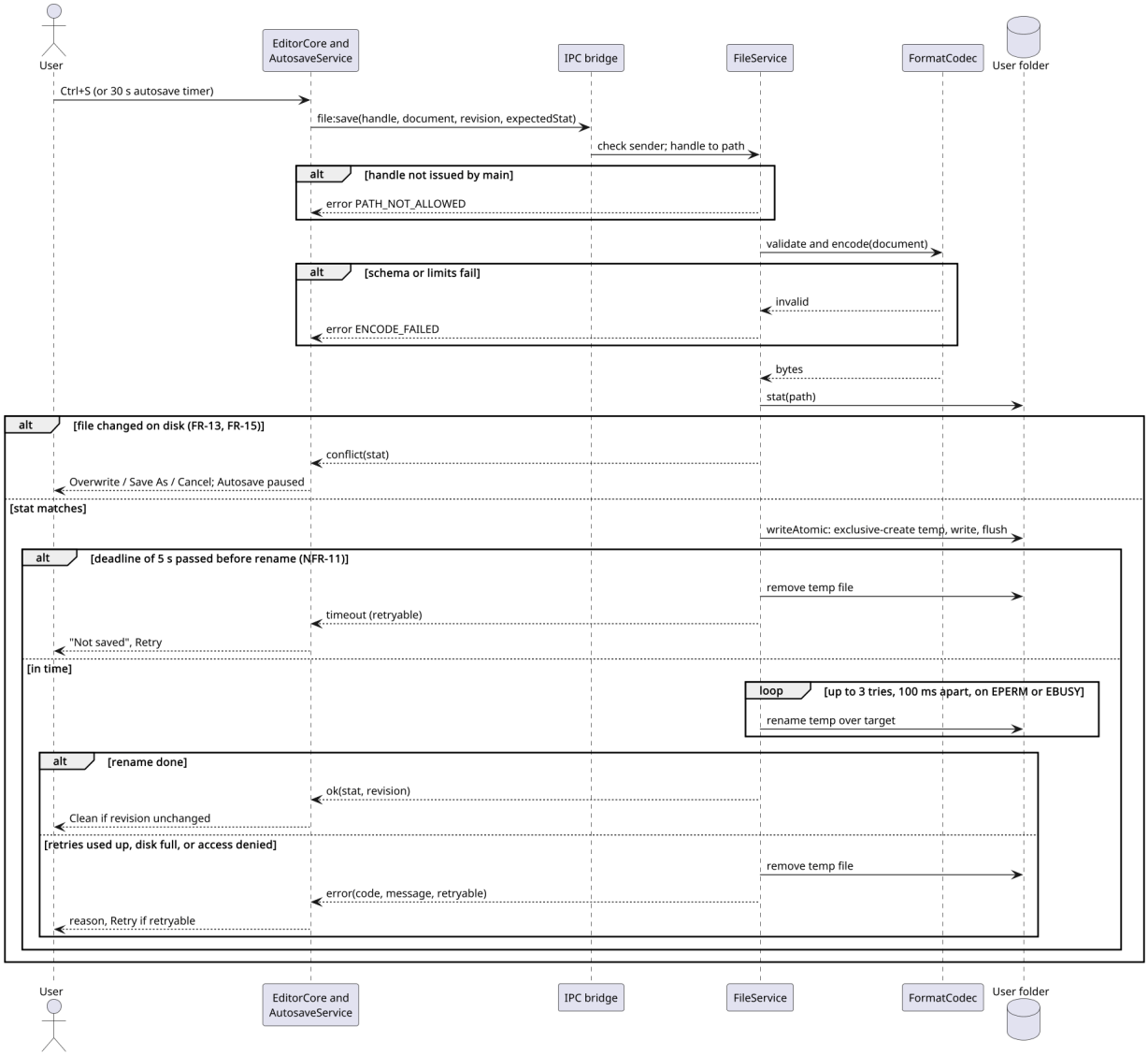


Figure 3. *Save and autosave, with conflict, timeout, and error*

Purpose: Shows the critical write path and its failure branches.

Assumption: The target is on a local NTFS disk; main finds the path from the tab handle.

Limitation: Save As and the first Save add the Save dialog and the replace prompt before this flow.

5.4 Sequence: export to Word

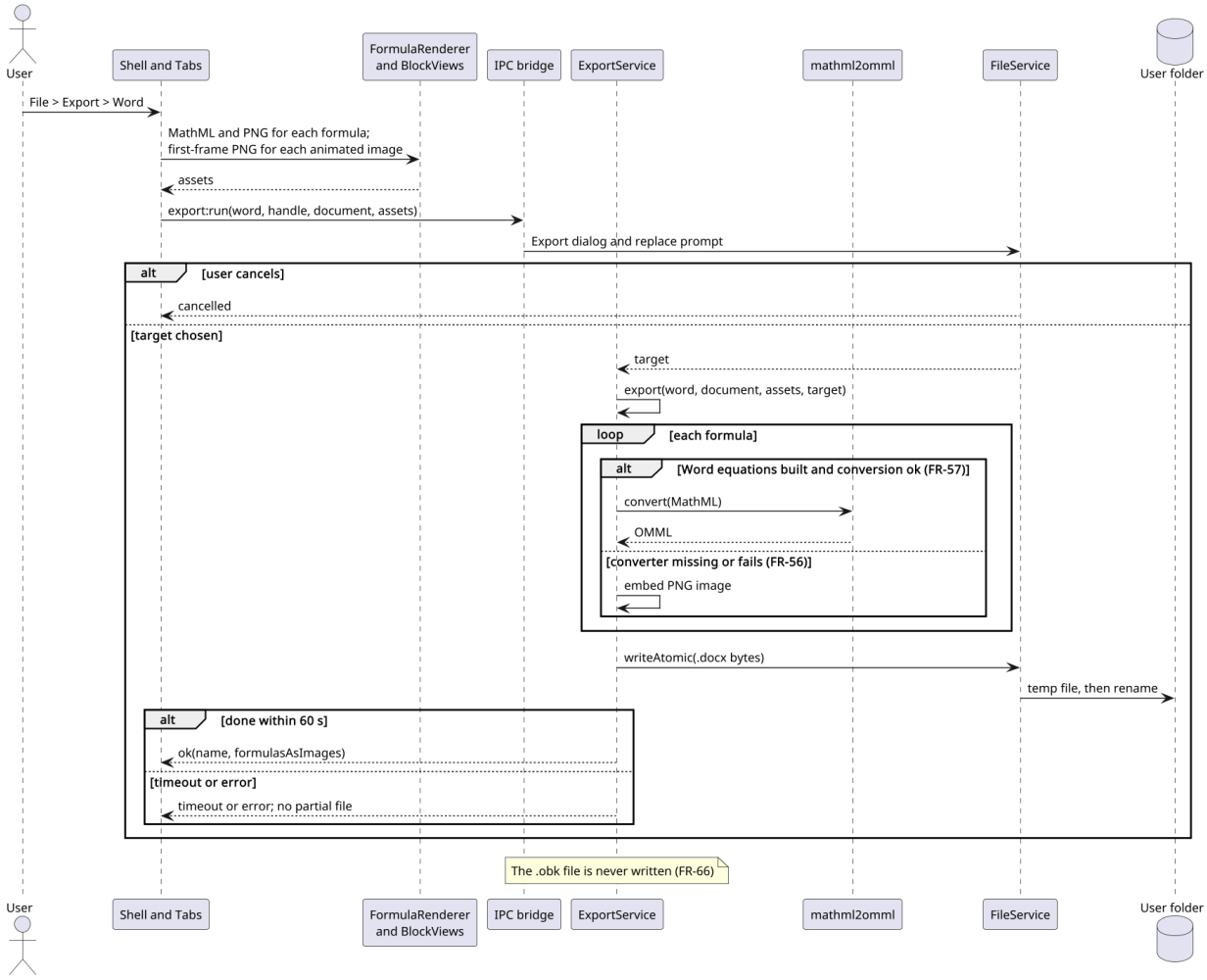


Figure 4. *Word export with equation fallback*

Purpose: Shows how formulas reach Word as equations or images, through the IPC bridge.

Assumption: KaTeX MathML and PNGs are made in the renderer, where KaTeX runs.

Limitation: Other blocks are mapped per the SRS Export reference; other results are in section 6.3.

5.5 Class view

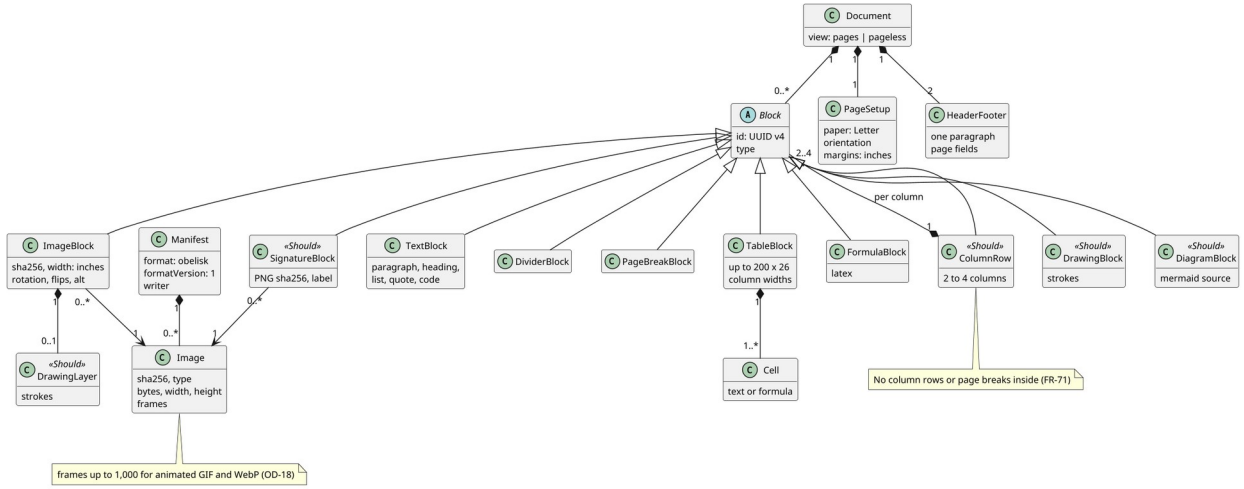


Figure 5. *Document domain model*

Purpose: Shows the manifest, the document, and every block kind stored in an .obk file.

Assumption: Format version 1 (OD-02); fields are in section 6.5.

Limitation: Lists, quotes, code, and headings are shown as TextBlock; inline items are not drawn.

5.6 Deployment view

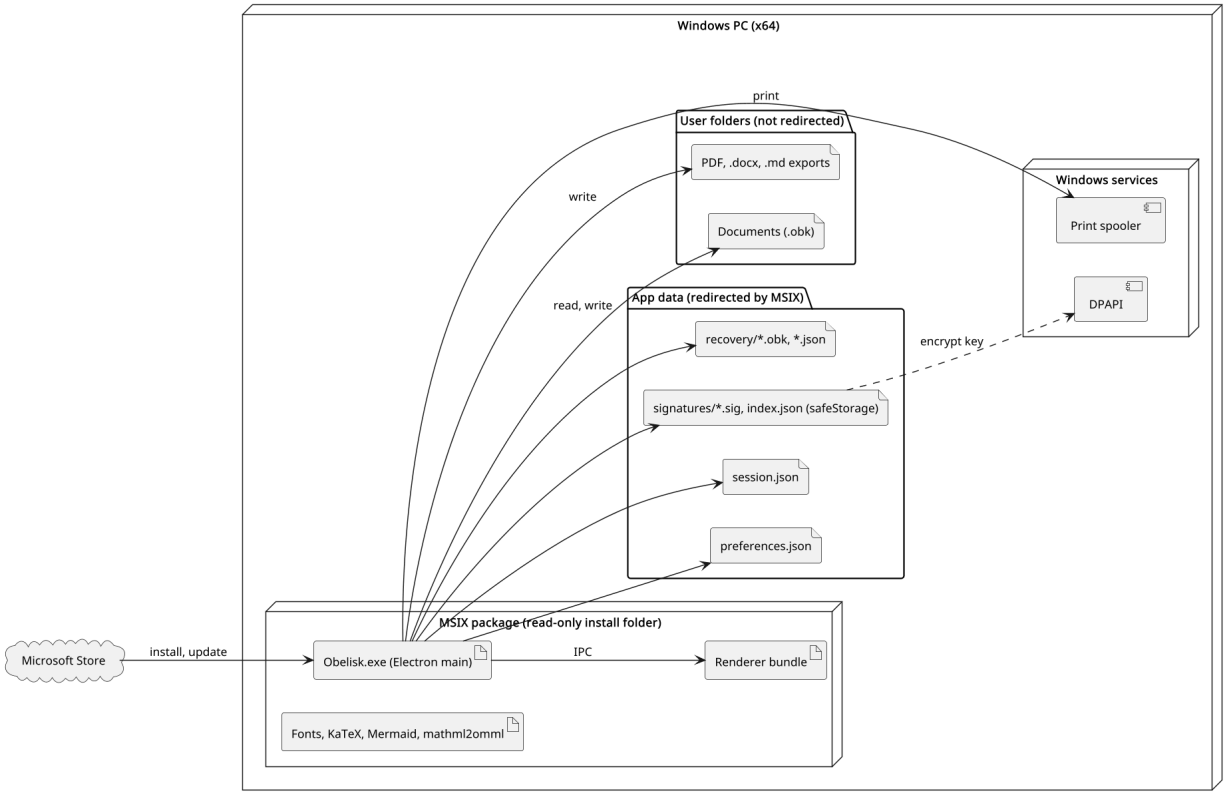


Figure 6. *Deployment on a Windows PC*

Purpose: Shows where each artifact lives and which folders MSIX redirects.

Assumption: Per-user MSIX install from the Store.

Limitation: MSIX redirection of app data is from Microsoft docs, not yet tested here.

UNVERIFIED: MSIX redirects writes to app data but not to user folders such as Documents. Check in spike S-01; source: Microsoft MSIX docs.

5.7 Screen flow

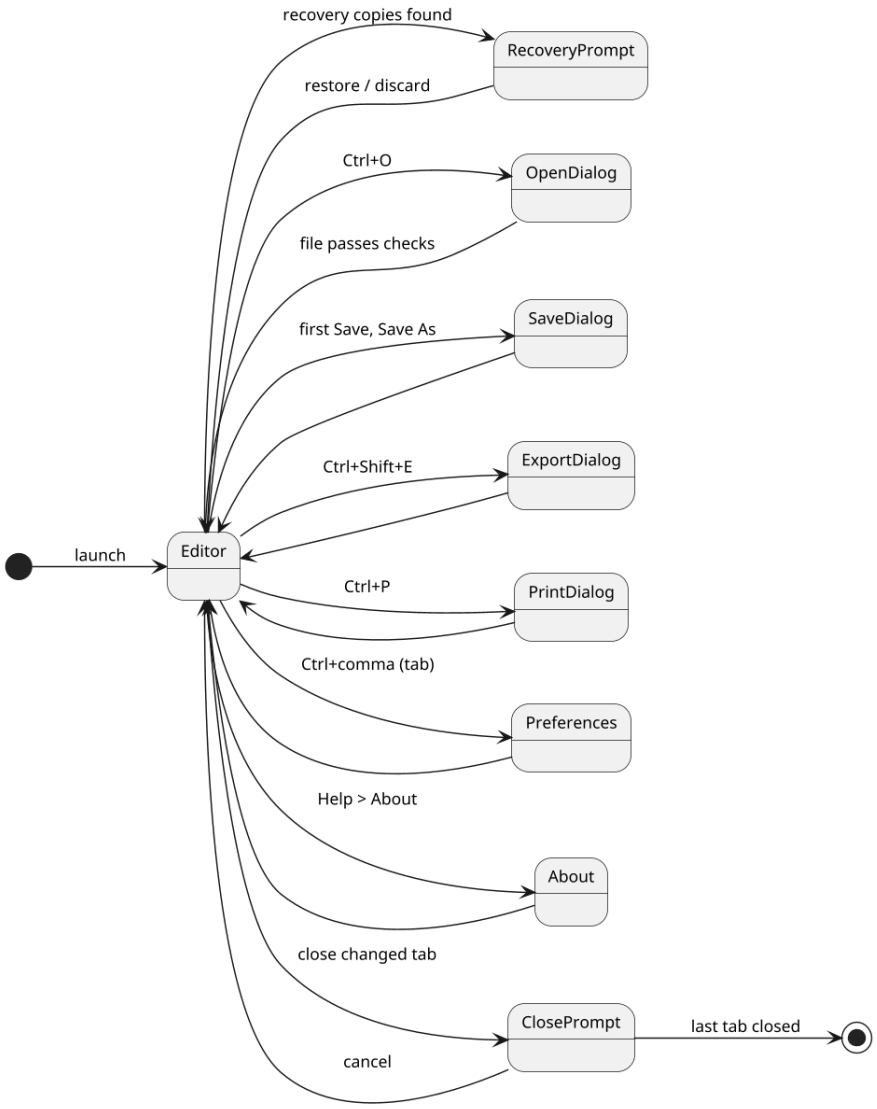


Figure 7. *Screen flow*

Purpose: Shows how the user moves between the editor, dialogs, and tabs.

Assumption: Preferences and About open as tabs, not windows.

Limitation: Menus and context menus are not drawn.

5.8 State machine: document

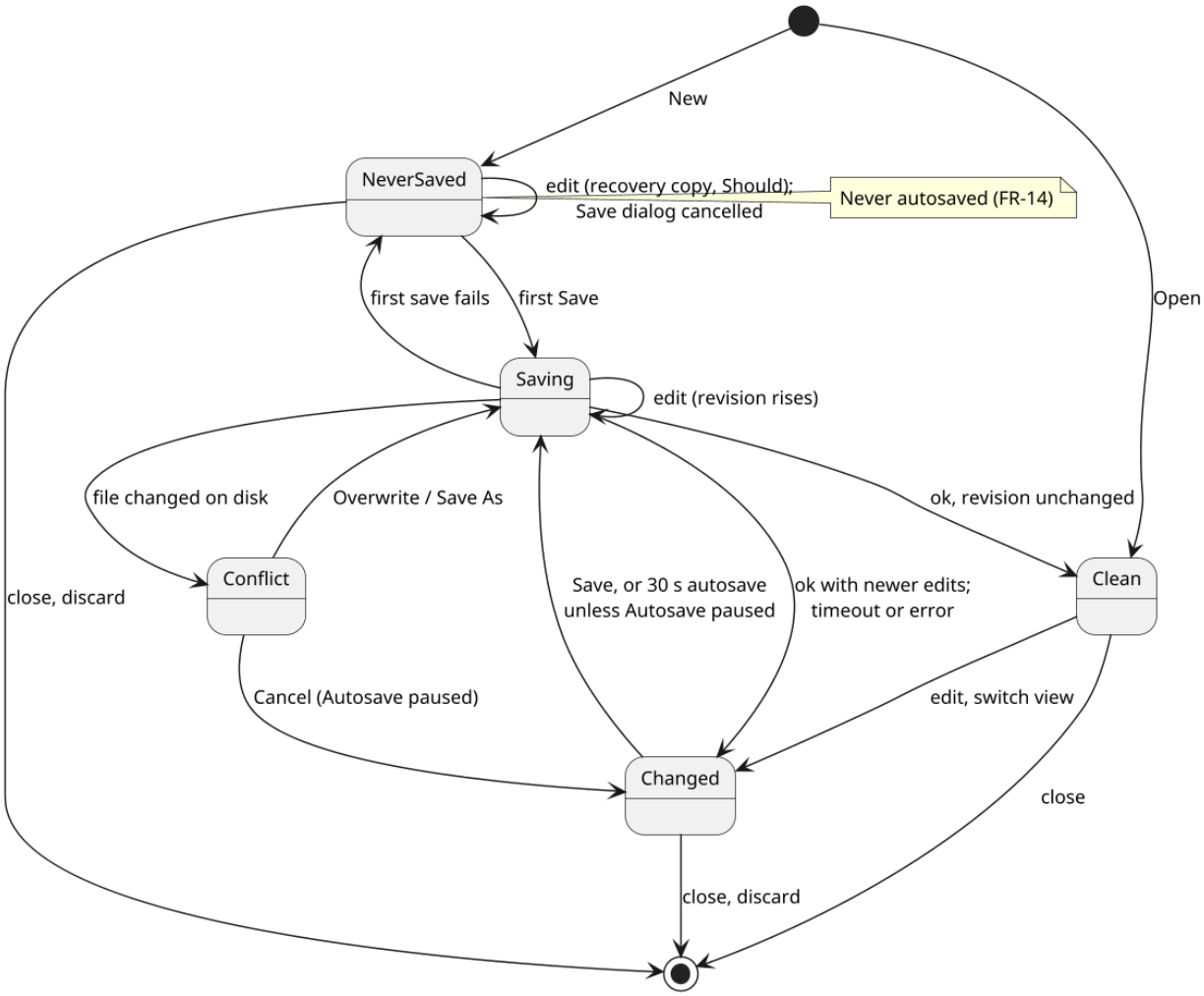


Figure 8. *Document save states*

Purpose: Shows the save state of one tab and when autosave applies.

Assumption: One tab per document; edits during a save are allowed.

Limitation: Recovery copies are a NeverSaved self-transition; Autosave paused is a flag, not a state.

6. Behavior and contract

The critical operation is saving a document (FR-10, FR-11, FR-14). Figure 3 shows the flow. Sections 6.5 to 6.10 set the file format and the checks on untrusted input.

6.1 Save contract

Field	Rule
Operation	file:save, called by Save, Save As, and the renderer's autosave timer; one path (ADR-002)
Inputs	Tab handle; document model; revision number; expected stat (size and modified time) or none; replaceConfirmed
Authorization	Main maps the handle to a path it issued from a dialog, an activation, or the session. Any other handle returns PATH_NOT_ALLOWED
Validation	Main checks the document against the schema and limits (sections 6.5 and 6.6) before encoding; a failure returns ENCODE_FAILED and writes nothing
Before writing	stat the target. A changed stat returns conflict. A new target that exists without replaceConfirmed returns exists
Atomicity	writeAtomic: a temp file `.~obk-<128-bit hex>.tmp` in the same folder, made with exclusive create, then flush and one rename over the target
Timeout	5 s (NFR-11), checked just before the rename. Past the deadline, the temp file is removed and nothing is renamed. A rename that has started finishes and returns ok
Retry	On EPERM or EBUSY during the rename (antivirus, indexer): 3 tries, 100 ms apart, then EBUSY_EXHAUSTED
Outputs	ok with the new stat and the saved revision; conflict with the current stat; exists; timeout (retryable); error with code, message, and retryable
Error codes	ENOSPC, EBUSY_EXHAUSTED (retryable); EACCES, ENOENT, PATH_NOT_ALLOWED, ENCODE_FAILED (not retryable)
State changes	Saving to Clean on ok only if the revision is unchanged, else Changed. Timeout or error: back to Changed, or NeverSaved after a failed first save
Autosave	Same call with the tab handle. On conflict it shows the choice once, pauses autosave for that tab, and the status bar shows Autosave paused. It resumes after the next successful Save, Overwrite, or Save As
Cleanup	On launch and after each save, only files matching the temp pattern and older than 10 minutes are removed (NFR-14)

UNVERIFIED: That fs.rename over an existing file maps to an atomic replace (MoveFileEx with replace) on NTFS from an MSIX-packaged app. Check in spike S-01; source: Node.js fs docs and libuv.

6.2 Alternatives

Case	Trigger	Result
Conflict	Another program changed the file	Return conflict; the user picks Overwrite (save with the new stat), Save As, or Cancel
Exists	A new target exists and the replace was not confirmed	Return exists; write nothing; file:saveAs shows the replace prompt and calls again after Replace
Timeout	No completion within 5 s (slow or network folder)	Return timeout; tab stays changed; status bar says Not saved; Retry offered

Case	Trigger	Result
Error	Disk full, access denied, path gone, retries used up, document fails the schema	Return error with its code; temp file removed; Retry offered when retryable
Not allowed	A handle main did not issue, or one for a closed tab	Return error PATH_NOT_ALLOWED; nothing is read or written
Cancelled	The user cancels the Save dialog or the replace prompt	Return cancelled; a never-saved tab stays NeverSaved

6.3 IPC surface

Channel	Direction	Payload	Results
`file:open`	R to M	None; main shows the Open dialog	ok with handle, name, document, stat; cancelled; refused with reason and message; error; alreadyOpen with tabId
`file:save`	R to M	Handle, document, revision, expected stat	ok, conflict, exists, timeout, error (section 6.1)
`file:saveAs`	R to M	Handle or none, document, revision	Main shows the Save dialog and replace prompt, then saves: file:save results with a new handle and name, or cancelled
`file:activated`	M to R	One file:open result per file: double-click, second instance, or session restore	None
`session:set`	R to M	Handles in tab order, active handle	ok; main stores paths only while Restore tabs is on
`export:run`	R to M	Format, handle or none, document, assets: MathML, formula PNGs, first-frame PNGs	ok with name and formulasAsImages; cancelled; error; timeout after 60 s
`print:run`	R to M	Document, assets	ok; cancelled; error
`prefs:get`, `prefs:set`	R to M	Key, value	Values; refused for an unknown key or bad value
`sig:list`	R to M	None	ok with IDs and names only; unavailable with a message
`sig:get`	R to M	Signature ID	ok with the PNG; unavailable; notFound
`sig:save`	R to M	Name, PNG	ok with the new ID; unavailable; limitReached; refused with reason; error
`sig:delete`	R to M	Signature ID	ok; notFound
`recovery:put`	R to M	Recovery ID or none, document, first line	ok with the recovery ID; error
`recovery:list`	R to M	None	IDs, first lines, and times, from the metadata records only
`recovery:restore`	R to M	Recovery ID	file:open results, as a never-saved tab; refused offers Discard only
`recovery:drop`	R to M	Recovery ID	ok; notFound
`link:open`	R to M	Address	ok after the user confirms; cancelled; refused for a scheme off the allowlist

R to M: renderer to main. M to R: main to renderer. Every error carries a code, a message, and retryable. Handles and IDs are made by main and looked up in its own lists; the renderer never sends a path.

6.4 Security baseline

These settings follow the Electron security checklist.

- contextIsolation on, sandbox on, nodeIntegration off in every renderer, including the hidden window used for print and PDF.
- The app loads from its own app: protocol, not file://.
- Content Security Policy: default-src 'none'; scripts, styles, and fonts from app: only; images from app:, data:, and blob:; no connect, frame, or object sources.
- A session-wide request filter in main cancels every request that is not app:, data:, or blob:. No crash reporter or updater runs; the Store updates the app.
- Every permission request is denied. Navigation, new windows, and webview tags are blocked.
- Links: only http, https, and mailto, checked on load, paste, and Ctrl+K, and again in main before opening. The prompt shows the full address (OD-20).
- The IPC bridge exposes named functions only. Main checks that each message comes from the app's own frame, validates every argument, and never passes the raw event to the renderer.
- The renderer never sends a path. Main shows every file dialog, keeps a handle-to-path map, and makes every recovery and signature ID itself.
- Documents are decoded from JSON into the schema; no stored HTML is ever rendered. KaTeX and Mermaid output enters the page only through FormulaRenderer and BlockViews.
- Command-line arguments: only existing regular .obk files are taken, up to 20 per launch; other arguments are ignored.
- Electron fuses runAsNode and nodeCliInspect are off, checked in the build.
- Dependencies are pinned by a lockfile and audited in every build. An Electron security release ships to the Store within 30 days (target).

Library settings for untrusted formula and diagram source:

Library	Setting	Why
KaTeX	trust false, set explicitly	Keeps \href, \url, \includegraphics, and the HTML commands off
KaTeX	maxSize 50, maxExpand 1000	Caps rule and box sizes, and macro expansion
KaTeX	throwOnError true	The editor shows KaTeX's own message (FR-55)
KaTeX	strict handler that refuses htmlExtension	HTML extensions are errors
KaTeX	A new, empty macros object for each render	A \gdef in one formula cannot change another
Mermaid	securityLevel strict (OD-19)	HTML tags in diagram text are encoded; click handlers are off
Mermaid	startOnLoad false, htmlLabels false	Diagrams render only when BlockViews asks; labels are SVG text
Mermaid	secure list adds themeCSS, themeVariables, fontFamily, htmlLabels, dompurifyConfig, theme	An init directive in a diagram cannot change them
Mermaid	maxTextSize 50,000 and maxEdges left at their defaults	Caps the work per diagram

UNVERIFIED: Which style-src KaTeX output needs under this policy, and whether Mermaid strict output needs any. Check in spike S-04.

UNVERIFIED: Fuses other than runAsNode and nodeCliInspect, and Mermaid's security advisories. Check when versions are pinned.

UNVERIFIED: How printToPDF sets up its window, how the parser used for pasted HTML treats scripts, and how custom clipboard formats behave. Check in the chosen Electron version.

UNVERIFIED: Decoding .obk files in a sandboxed utilityProcess instead of main. Check in spike S-08.

6.5 Document schema

Format version 1. One machine-readable schema file in the repository is the single source for FormatCodec, the editor schema, and the Node tests (NFR-21). Unknown fields are refused (FR-09).

manifest.json

Field	Type	Rule
`format`	String	Must be obelisk
`formatVersion`	Integer	1 up to this app's version; newer is refused (section 6.7)
`writer`	String	App version, such as 1.0.0; information only
`images`	List	One item per image entry; may be empty
`images: sha256`	String	64 lowercase hex digits; matches the entry name and its bytes
`images: type`	png, jpeg, gif, webp	Matches the first bytes and the extension
`images: bytes`	Integer	1 byte to 20 MB; matches the real size
`images: width, height`	Integer	1 to 8000 pixels; matches the image header
`images: frames`	Integer	1 for still images; up to 1,000 for animated GIF and WebP

document.json, top level

Field	Type	Rule
`view`	pages or pageless	Required
`page`	Orientation, margins in inches	Required; text width and height at least 2 in
`header, footer`	One paragraph each	Text with marks and page-number fields only; may be empty
`blocks`	List of blocks	Required; may be empty; up to 20,000 blocks in all

Blocks

Block	Fields	Rule
paragraph	Inline content; alignment; line spacing; space before and after; indents	First-line and hanging indents are not both set
heading	Level 1 to 6; inline content	None
list	Bulleted, numbered, or checklist; items with level 0 to 5, checked, content	checked only in checklists

Block	Fields	Rule
quote, code	Inline content; code is plain text	Code has no marks
divider, pageBreak	No fields	pageBreak not inside column rows
table	Rows of cells; cell text or formula; column widths	Up to 200 by 26; cell up to 10,000 characters
image	sha256; width in inches; rotation 0, 90, 180, 270; flips; alt; drawing layer (Should)	sha256 is in the manifest
formula	LaTeX source	Up to 10,000 characters
columns (Should)	2 to 4 columns; shares; blocks per column	No column rows or page breaks inside
drawing (Should)	Strokes of points, width, color; size	Up to 2,000 strokes and 100,000 points
signature (Should)	PNG sha256; line; width; label lines; alignment; blank mode	Label up to 100 characters per line, 2 lines
diagram (Should)	Mermaid source	Up to 50,000 characters

Inline content

Item	Fields	Rule
text	Text and marks: bold, italic, underline, strike, code, font, size, color	Up to 100,000 characters per block
link	Address and text	http, https, or mailto; up to 2,048 characters
inlineFormula	LaTeX source	Up to 10,000 characters
dateChip (Should)	Date and time as YYYY-MM-DDTHH:MM, no zone; format	Static; never updates on its own
pageField	page or pages; number format	Header and footer only

Units and IDs

Item	Rule
Lengths	Inches as decimal numbers, up to 3 places
Font size	Points, one of the sizes in the SRS Formatting reference, 8 to 72
Colors	#RRGGBB
Strokes	Points in the block's own pixel space
IDs	Each block has an id: a random version 4 UUID in lowercase, unique in the file. Paste and duplicate make new IDs; a repeated ID is refused

A block marked Should is removed from format 1 before release if its feature is cut, so no released file holds it.

6.6 File checks and limits

Item	Limit
File on disk	200 MB
Archive entries	2,000
document.json, uncompressed	25 MB (OD-21)

Item	Limit
Each image, uncompressed	20 MB
All entries, uncompressed	250 MB
Image size	8000 by 8000 pixels, read from the header before decoding
Animated image	1,000 frames; 250 million pixels across all frames; each frame inside the image size
Blocks, text, tables, formulas, diagrams, drawings	As in section 6.5
Font family name; alt text	1 to 64 characters; up to 500 characters

Limits are targets, to tune against NFR-03 and NFR-02. Caps count the bytes actually read, not the sizes the archive headers claim.

- Entries are read as streams in memory and never unpacked to disk. A read stops as soon as a cap is passed.
- Only stored and deflate entries. Encrypted entries, multi-disk archives, and nested archives are refused. Central and local headers must agree.
- Only the exact names manifest.json, document.json, and images/ followed by 64 hex digits and .png, .jpg, .gif, or .webp. Folders, repeated names, and links are refused. This rules out backslashes, drive letters, network paths, and case duplicates.
- JSON: duplicate keys, unpaired surrogates, and numbers that are not finite are refused; a byte order mark is ignored. Writes use a fixed key order, so the same document gives the same bytes.
- Images: type from the first bytes; width, height, and frames from the header before decoding; SVG refused. Pixels are decoded only in the sandboxed renderer.
- Animated GIF and WebP: FormatCodec walks the GIF blocks or WebP chunks to count frames and read each frame's size, without decoding pixels. A file over a limit is refused (OD-18).
- The same checks run on open, on recovery and session restore, on pasted Obelisk blocks, and on every document main receives to save.

Rule	Where enforced	How verified
Archive caps and entry allowlist	FormatCodec, on open and restore	H-01 to H-09
Schema, unknown fields, duplicate keys, IDs	FormatCodec; EditorCore schema	H-10 to H-15
Image type, size, and header	FormatCodec; ClipboardBridge on paste	H-16 to H-19
Animated frames and pixel budget	FormatCodec; ClipboardBridge on paste	H-27 to H-29
Link schemes	EditorCore; main before opening	H-20, H-21
Formula and diagram settings	FormulaRenderer; BlockViews	H-22 to H-24
Handles and IDs from main only	IPC bridge; FileService; RecoveryService; SignatureStore	H-25, H-30
Export escaping and names	ExportService	H-26
Stored text is never markup	EditorCore; FormatCodec	H-31
App data checked on read	PreferencesStore; RecoveryService; SignatureStore	H-32

UNVERIFIED: Counting GIF and WebP frames without decoding, and making the first frame in the renderer. Check in spike S-06.

6.7 Format versions

- formatVersion is a whole number; each schema change raises it by 1. Format 1 is the first, so no conversion runs in version 1.0 of the app.
- An app opens every format up to its own, converting older ones in memory one step at a time. Each step is a pure function in FormatCodec with Node tests.
- A newer file is refused with "Made by a newer Obelisk; update it from the Microsoft Store" (FR-09).
- A converted document opens with autosave paused. The first save asks "Saving updates this file; older versions of Obelisk can't open it." After the user agrees, autosave resumes (OD-14).
- Within one format version, unknown fields are still refused.

6.8 App data records

All records live in the app data folder and are written through writeAtomic.

Record	File	Contents	Checks on read
Preferences	`preferences.json`	version and each preference	A bad or unknown value uses the default for that key; the app always starts
Session	`session.json`	version, paths in tab order, active tab	Kept only while Restore tabs is on. Each path reopens through the file:open checks; missing files are named in one message (OD-17)
Saved signature	`signatures/<uuid>.sig`	Name, time, and PNG, encrypted as a whole	Decrypted only by sig:get, never to a temp file. Re-encrypted when safeStorage asks. Unavailable: nothing shown, nothing deleted (OD-15)
Signature index	`signatures/index.json`	IDs and order only, no names	A bad entry is skipped; the others still load
Recovery copy	`recovery/<uuid>.obk` and `.json`	A format 1 .obk; times and the first line, up to 80 characters	Restore runs the FormatCodec checks; a failing copy offers Discard only. Kept and offered at every launch until restored or discarded (OD-16)
Temp save file	`.~obk-<hex>.tmp` beside the target	Partial bytes of a save	Made with exclusive create; cleanup removes only this pattern (NFR-14)

UNVERIFIED: safeStorage inside an MSIX package, including the temporarily unavailable state. Check in spike S-07.

6.9 Paste, export, and print

- Pasted HTML is parsed into a detached document and mapped through an allowlist into the schema; raw HTML never enters the page. Pasted Obelisk blocks pass the file checks and get new IDs. Pasted images follow the image limits; SVG is refused.
- Copy builds HTML from the schema and escapes it; it never copies the live page.
- One export interface, export(format, document, assets, target), reached only through export:run (ADR-006). Main shows the Export dialog and the replace prompt.
- Every export file goes through writeAtomic. A new Markdown images folder is written as a temp folder, then renamed into place. An existing one is named in the replace prompt; each image in it is replaced through writeAtomic and other files are left alone.

- Markdown escapes text, alt text, and link text; formulas are written only as LaTeX between dollar signs; no raw HTML. Images are named image-001.png and so on. Animated GIF and WebP are copied as their original, already checked, bytes.
- Word text goes through the library's text functions only. Word, PDF, and print use the first frame of an animated image: a PNG made in the renderer at export time, checked by main, never stored.
- PDF and print use one hidden window with the settings in section 6.4.
- An export stops after 60 s and returns timeout. The temp file or folder is removed, and a replaced file keeps its old bytes (NFR-04 sets the 10 s target).

6.10 Hostile sample set

Each sample runs under Node against FormatCodec and, where marked by the case, in the packaged app. No sample may change the file it opens.

ID	Case	Expected
H-01	File over 200 MB; archive with 2,001 entries	Refused before reading entries
H-02	Entry that inflates past its cap (zip bomb), with a small declared size	Read stops at the cap; refused
H-03	Encrypted entry; unknown compression method; multi-disk archive	Refused
H-04	Entry names with ../, a leading slash, a backslash, or a drive letter	Refused; nothing written
H-05	Entry names with a network path, a stream suffix, or a device name such as CON	Refused
H-06	Two names that differ only by case; a repeated name	Refused
H-07	Central and local headers that disagree	Refused
H-08	A folder entry, a link entry, or a nested .zip	Refused
H-09	Missing manifest.json or document.json	Refused
H-10	Unknown field; wrong type; value out of range	Refused
H-11	Duplicate JSON keys; unpaired surrogate; NaN or Infinity	Refused
H-12	Repeated block ID	Refused
H-13	Over 20,000 blocks; table over 200 by 26; text over its cap	Refused
H-14	Columns inside columns; a page break inside a column row	Refused
H-15	formatVersion newer than the app	Refused with the update message
H-16	Image whose first bytes do not match its type or extension	Refused
H-17	Image header over 8000 by 8000 pixels	Refused before decoding
H-18	SVG entry, or SVG pasted	Refused; dropped from paste
H-19	Image entry not in the manifest, or a hash that does not match	Refused
H-20	Link with javascript:, file:, data:, or a custom scheme	Refused on open; dropped on paste
H-21	Pasted HTML with script, event handlers, remote images, and iframes	Nothing runs; no request is sent
H-22	Formula with \href, \url, \includegraphics, or HTML commands	Shown as an error
H-23	Formula with a huge size or deep macro expansion; a \gdef carried to another formula	Error; other formulas unchanged
H-24	Diagram with HTML in labels, a click handler, or an init directive	Encoded or ignored; nothing runs
H-25	Command line with flags, a folder, or 21 files	Flags ignored; only .obk files; first 20 opened
H-26	Export of text, alt text, and link text with Markdown and XML markup	Escaped; file opens cleanly
H-27	Animated GIF with 1,001 frames	Refused on open and paste

ID	Case	Expected
H-28	Animated WebP whose frames add up to over 250 million pixels	Refused
H-29	Animated GIF with a frame outside the image size	Refused
H-30	Fake renderer: unknown handle, a recovery ID that names another file	PATH_NOT_ALLOWED or notFound; nothing touched
H-31	A paragraph whose text holds script and remote-image tags	Opens; shown as text; no request
H-32	A damaged recovery copy, session.json, preferences.json, or signature index	Discard offered or defaults used; nothing deleted

7. Architecture decisions

Decisions are based on SRS 1.1. They are Proposed until accepted; the two marked Accepted follow decisions made on October 23, 2026. Structure: ADR-001. Failure handling: ADR-002.

ADR	Decision	Status
ADR-001	The main process owns all file access	Proposed
ADR-002	Save, Save As, and autosave share one save call and one atomic-write helper	Proposed
ADR-003	One ZIP container per document	Proposed
ADR-004	Editor built on a ProseMirror-based core with custom block views	Proposed
ADR-005	Formulas: KaTeX on screen and PDF, OMML for Word, image fallback	Proposed
ADR-006	Each exporter is a separate module behind one interface	Proposed
ADR-007	Saved signatures encrypted with safeStorage in app data	Proposed
ADR-008	Microsoft Store MSIX as the only release channel	Proposed
ADR-009	Single instance with file activation, and recovery copies in app data	Proposed
ADR-010	Diagrams render in the renderer with Mermaid at securityLevel strict	Accepted 2026-10-23
ADR-011	Older format versions convert forward; newer ones are refused	Accepted 2026-10-23

ADR-001 The main process owns all file access

Field	Content
Status	Proposed
Context	Untrusted files and pastes reach the renderer (NFR-06, NFR-07). Disk access must be in one place (NFR-22).
Decision	Only main-process services touch the disk. Renderers call named functions on the IPC bridge and never send paths: main shows the dialogs and hands out handles. Main checks the sender and validates every document before encoding.
Alternatives	Node in the renderer: simpler, but any script bug could reach the disk. A local server: needless for one user.
Consequences	Smaller attack surface and one place to test. Costs: IPC types to keep in step, and copying the document across IPC on save.
Revisit when	Saving very large documents across IPC misses NFR-03.

ADR-002 Save, Save As, and autosave share one save call and one atomic-write helper

Field	Content
Status	Proposed
Context	A file must never be left half-written (NFR-10 to NFR-14). Autosave runs often without the user watching (FR-14, FR-15).
Decision	file:save checks the disk stat, then calls FileService.writeAtomic: an unpredictable temp name in the same folder, exclusive create, flush, one rename. Save, Save As, and the

Field	Content
	renderer's autosave timer all call file:save. Exports, recovery, signatures, and preferences use writeAtomic too.
Alternatives	Write in place: simpler, but a crash leaves part of a file. Autosave to a side copy: two stores to keep in step.
Consequences	One helper to test under Node (NFR-21); autosave inherits the conflict and timeout rules. Costs: a temp file per save and leftover cleanup (NFR-14).
Revisit when	Saves to network or cloud-synced folders show failures.

ADR-003 One ZIP container per document

Field	Content
Status	Proposed
Context	A document holds blocks, images, and settings, and must be complete on its own (FR-49, FR-17).
Decision	An .obk file is a ZIP with manifest.json, document.json, and an images folder of hash-named files, per sections 6.5 and 6.6.
Alternatives	One JSON file with base64 images: simple, but large and slow. A folder per document: easy to break by moving files.
Consequences	Images stay compact and files are easy to inspect. Costs: size caps, an entry allowlist, and schema checks against hostile archives (OD-21); older formats convert forward (OD-14).
Revisit when	Format version 2 is needed (OD-02).

ADR-004 Editor built on a ProseMirror-based core with custom block views

Field	Content
Status	Proposed
Context	Blocks, typed shortcuts, undo, and schema checks are core needs (FR-32, FR-35, FR-37).
Decision	Use a ProseMirror-based core with a strict schema. Each block kind is a custom node view in BlockViews.
Alternatives	contenteditable from scratch: too much work for one developer. Other editor frameworks: less control over the schema.
Consequences	A proven model for undo and input rules. Costs: learning curve; pagination is custom (FR-74).
Revisit when	Pages view layout cannot meet NFR-02.

ADR-005 Formulas: KaTeX on screen and PDF, OMML for Word, image fallback

Field	Content
Status	Proposed
Context	Formulas are the standout feature (FR-53). Word needs native equations (FR-57), but conversion may fail (FR-56).

Field	Content
Decision	KaTeX renders on screen and in PDF, with the fixed settings in section 6.4. For Word, KaTeX MathML goes through mathml2omml; on failure, or if not built, the formula is a PNG.
Alternatives	MathJax: heavier. Images only: Word users cannot edit equations.
Consequences	Word export never fails because of a formula. Costs: an LGPL module (NFR-29) and possible fidelity gaps (R-01).
Revisit when	mathml2omml fidelity is poor on the SRS samples.

ADR-006 Each exporter is a separate module behind one interface

Field	Content
Status	Proposed
Context	PDF, Word, and Markdown map every block differently (SRS Export reference). HTML and .txt may come later.
Decision	ExportService calls one interface, export(format, document, assets, target), reached only through export:run. Each format is a module testable under Node, except PDF, which uses printToPDF. Files go through writeAtomic.
Alternatives	One exporter with format switches: grows tangled.
Consequences	Formats are tested and cut on their own; HTML can be added later. Cost: some shared mapping code.
Revisit when	A future format needs a different input than the block tree.

ADR-007 Saved signatures encrypted with safeStorage in app data

Field	Content
Status	Proposed
Context	Signatures are sensitive (NFR-08, NFR-09). The app has no account or key store.
Decision	SignatureStore keeps each signature as a PNG in its own file, encrypted with the async safeStorage API (OD-15). sig:list returns names only; sig:get decrypts one. Unavailable or temporarily unavailable: save and show nothing, delete nothing, say so.
Alternatives	Plain files: readable by any user with access. Own key file: the key sits next to the data.
Consequences	Uses DPAPI with no key handling. Limitation: DPAPI guards against other Windows users, not other apps of the same user (Electron safeStorage docs).
Revisit when	Electron changes the safeStorage API again.

ADR-008 Microsoft Store MSIX as the only release channel

Field	Content
Status	Proposed
Context	Release is Store only (NFR-24, NFR-31, NFR-27).
Decision	Package as MSIX, per-user, with the .obk file-type association; updates only through the

Field	Content
	Store.
Alternatives	A separate installer: needs code signing and an updater.
Consequences	Free signing and updates; clean uninstall. Costs: certification time (R-04) and MSIX file-system rules (R-06).
Revisit when	Users ask for a download outside the Store.

ADR-009 Single instance with file activation, and recovery copies in app data

Field	Content
Status	Proposed
Context	Double-clicked files open as tabs (FR-18). Never-saved tabs need crash safety (FR-19 to FR-21).
Decision	Main takes the single-instance lock; a second launch passes its arguments through the second-instance event, and main keeps only .obk files. RecoveryService writes recovery copies through writeAtomic, named by UUIDs main makes; the renderer's timer sends them (OD-16).
Alternatives	Many windows, one per file: breaks the tab model. Recovery in the user's folder: leaves files the user never asked for.
Consequences	One window, one place for recovery. Costs: the lock must release on crash, and the activation path needs an MSIX test (R-06).
Revisit when	Users want one window per document.

ADR-010 Diagrams render in the renderer with Mermaid at securityLevel strict

Field	Content
Status	Accepted 2026-10-23
Context	Diagram sources come from untrusted files (FR-82). Mermaid output is SVG placed in the page.
Decision	BlockViews sets Mermaid up once with securityLevel strict, startOnLoad false, htmlLabels false, and an extended secure list (section 6.4) (OD-19).
Alternatives	sandbox level: renders in a frame, whose fit with the policy, print, and PDF is unchecked. loose or antiscript: allow HTML in labels.
Consequences	Labels are plain text and click handlers are off. Cost: some styling a diagram could set is ignored.
Revisit when	Mermaid changes its security settings, or a needed diagram type requires HTML labels.

ADR-011 Older format versions convert forward; newer ones are refused

Field	Content
Status	Accepted 2026-10-23
Context	Files outlive app versions, and unknown fields are refused within a version (FR-09,

Field	Content
	OD-14).
Decision	FormatCodec opens every format up to its own through tested one-step conversions, refuses newer formats, and asks before the first save upgrades a file (section 6.7).
Alternatives	Refuse every other version: simplest, but each schema change cuts off old files. Ignore unknown fields: weakens the hostile-file checks.
Consequences	Old files keep opening and users are told before an upgrade. Cost: conversion code kept for every past version.
Revisit when	A change cannot be written as a one-way conversion.

8. Risks

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-01	mathml2omml output differs from KaTeX for some formulas	Medium	Medium	Test the SRS formula samples in Word early; the image fallback covers failures (FR-56)	Developer
R-02	LGPL terms for mathml2omml inside the MSIX are not met	Low	High	Ship it unbundled as its own module, with source link and license text (NFR-29); check the LGPL text	Developer
R-03	Column rows split across pages differently on screen and in PDF	High	Medium	One layout module used by Pages view and print; PDF compare tests (FR-74)	Developer
R-04	Store certification is delayed near release	Medium	High	Submit a private test build one month before release (NFR-31); allow certification time in security updates	Developer
R-05	Some LibreOffice fonts are not OFL	Medium	Medium	Check each license file before bundling; drop any font that does not fit (OD-08)	Developer
R-06	MSIX file activation or file writes behave differently than expected	Medium	Medium	Spikes S-01 and S-02 in a packaged build before the file features are built	Developer
R-07	Antivirus or indexers lock the target during replace	Medium	Medium	Retry 3 times at 100 ms, then report; never leave a partial file (NFR-10)	Developer
R-08	A Chromium, KaTeX, or Mermaid bug lets a crafted file run code in a renderer	Low	High	Sandbox and policy (section 6.4); main refuses what it did not issue; security releases within 30 days (QS-07)	Developer

9. Technology stack

Layer	Technology	Version	Why	License
Desktop runtime	Electron	UNVERIFIED (45 or later)	Chromium layout and printToPDF; one codebase	MIT

Layer	Technology	Version	Why	License
Interface	React, TypeScript	UNVERIFIED	Common and typed	MIT; Apache-2.0
Editor core	ProseMirror-based library (proposal)	UNVERIFIED	Schema, undo, input rules	MIT
Math	KaTeX	UNVERIFIED	Fast, offline, MathML output	MIT
Word math	mathml2omml	0.5.0 (npm)	MathML to OMML	LGPL-3.0-or-later
Word file	A .docx writer library (proposal)	UNVERIFIED	Builds .docx under Node	UNVERIFIED
PDF and print	Electron printToPDF and print	With Electron	Same layout as the screen	MIT
Diagram	Mermaid, securityLevel strict	UNVERIFIED	Text diagrams, offline	MIT (UNVERIFIED)
ZIP	A ZIP library for Node (proposal)	UNVERIFIED	Reads and writes .obk	UNVERIFIED
Tests	A Node test runner with coverage (proposal)	UNVERIFIED	NFR-21, NFR-23	UNVERIFIED
Packaging	MSIX tooling for Electron (proposal)	UNVERIFIED	Store package	UNVERIFIED

UNVERIFIED: Versions and the licenses marked UNVERIFIED. Fix them after the spikes below, from each project's package page.

9.1 Spikes

Spike	Question	Settles
S-01	In a packaged MSIX build, save to Documents and confirm the write is not redirected and the rename replaces the file in one step.	R-06, R-07
S-02	Double-click an .obk file with Obelisk running; confirm the path arrives through the second-instance event.	R-06
S-03	Convert the SRS formula samples with mathml2omml and open them in Word.	R-01
S-04	Render a Mermaid flowchart at securityLevel strict in a sandboxed renderer with the network off, and find the style-src KaTeX and Mermaid output need.	OD-05, OD-19
S-05	Confirm whether the Windows Save dialog asks before replacing a file, as Electron sets it.	FR-12
S-06	Count frames and read frame sizes of animated GIF and WebP files without decoding, and make a first-frame PNG in the renderer.	OD-18
S-07	In a packaged MSIX build, encrypt and decrypt a signature with the async safeStorage API, and force the unavailable and re-encrypt states.	OD-15
S-08	Decode .obk files in a sandboxed Electron utilityProcess instead of main.	R-08

10. Traceability

Every Must FR maps to at least one component. Package-wide requirements map to the build and release process.

Requirement	Priority	Component	ADR	Verification
FR-01	Must	Shell and Tabs	-	Demo
FR-02	Must	Shell and Tabs	-	Inspection
FR-03	Must	Shell and Tabs	-	Test
FR-04	Must	Shell and Tabs	-	Test
FR-05	Must	Shell and Tabs	-	Inspection
FR-06	Must	PreferencesStore	-	Test
FR-07	Must	Shell and Tabs	-	Test
FR-08	Must	FileService	-	Test
FR-09	Must	FormatCodec	ADR-011	Test
FR-10	Must	FileService	-	Test
FR-11	Must	FileService	-	Demo
FR-12	Must	FileService	-	Test
FR-13	Must	FileService	-	Test
FR-14	Must	AutosaveService, FileService	ADR-002	Test
FR-15	Must	AutosaveService, FileService	ADR-002	Test
FR-16	Must	Shell and Tabs	-	Test
FR-17	Must	FormatCodec	ADR-003	Test
FR-18	Should	FileService	ADR-009	Demo
FR-19	Should	AutosaveService, RecoveryService	ADR-009	Inspection
FR-20	Should	Shell and Tabs	-	Test
FR-21	Should	RecoveryService	ADR-009	Inspection
FR-22	Must	EditorCore	-	Test
FR-23	Must	Shell and Tabs	-	Test
FR-24	Must	EditorCore	-	Test
FR-25	Must	EditorCore	-	Test
FR-26	Must	EditorCore	-	Test
FR-27	Must	EditorCore	-	Inspection
FR-28	Must	EditorCore	-	Test
FR-29	Must	EditorCore	-	Test
FR-30	Must	EditorCore	-	Test
FR-31	Must	EditorCore	-	Demo
FR-32	Must	EditorCore	ADR-004	Test
FR-33	Must	EditorCore	-	Demo
FR-34	Must	EditorCore	-	Test
FR-35	Must	EditorCore	ADR-004	Test
FR-36	Must	ClipboardBridge	-	Test
FR-37	Must	EditorCore	ADR-004	Test
FR-38	Must	EditorCore	-	Test

Requirement	Priority	Component	ADR	Verification
FR-39	Must	ClipboardBridge	-	Test
FR-40	Must	EditorCore	-	Test
FR-41	Must	ClipboardBridge	-	Test
FR-42	Must	EditorCore	-	Test
FR-43	Must	EditorCore	-	Test
FR-44	Must	EditorCore	-	Inspection
FR-45	Must	BlockViews	-	Test
FR-46	Must	BlockViews	-	Test
FR-47	Must	BlockViews	-	Test
FR-48	Must	BlockViews	-	Test
FR-49	Must	BlockViews	ADR-003	Test
FR-50	Must	BlockViews	-	Test
FR-51	Must	BlockViews	-	Test
FR-52	Must	BlockViews	-	Inspection
FR-53	Must	FormulaRenderer	ADR-005	Test
FR-54	Must	FormulaRenderer	-	Demo
FR-55	Must	FormulaRenderer	-	Test
FR-56	Must	ExportService	ADR-005	Test
FR-57	Should	ExportService	ADR-005	Test
FR-58	Must	EditorCore	-	Test
FR-59	Must	EditorCore	-	Test
FR-60	Must	EditorCore	-	Test
FR-61	Must	ExportService	-	Demo
FR-62	Must	ExportService	-	Test
FR-63	Must	ExportService	-	Test
FR-64	Must	ExportService	-	Test
FR-65	Must	ExportService	-	Test
FR-66	Must	ExportService	-	Test
FR-67	Should	BlockViews	-	Test
FR-68	Should	BlockViews	-	Test
FR-69	Should	BlockViews	-	Test
FR-70	Should	BlockViews	-	Test
FR-71	Should	BlockViews	-	Test
FR-72	Should	BlockViews	-	Test
FR-73	Should	BlockViews	-	Test
FR-74	Should	BlockViews	ADR-004	Test
FR-75	Should	BlockViews	-	Test
FR-76	Should	BlockViews	-	Test
FR-77	Should	ExportService	-	Inspection
FR-78	Should	BlockViews	-	Test
FR-79	Should	SignatureStore	-	Test

Requirement	Priority	Component	ADR	Verification
FR-80	Should	BlockViews	-	Inspection
FR-81	Should	ExportService	-	Test
FR-82	Should	BlockViews	ADR-010	Test
FR-83	Should	BlockViews	-	Test
NFR-01	Must	Build and release	-	Test
NFR-02	Must	EditorCore	ADR-004	Test
NFR-03	Must	FileService	ADR-001	Test
NFR-04	Must	ExportService	-	Test
NFR-05	Must	Build and release	-	Analysis
NFR-06	Must	ClipboardBridge	ADR-001	Test
NFR-07	Must	FormatCodec	ADR-001	Test
NFR-08	Should	SignatureStore	ADR-007	Inspection
NFR-09	Should	SignatureStore	ADR-007	Test
NFR-10	Must	FileService	ADR-002	Test
NFR-11	Must	FileService	-	Test
NFR-12	Must	FileService	-	Test
NFR-13	Must	AutosaveService	-	Test
NFR-14	Must	FileService	ADR-002	Test
NFR-15	Must	Shell and Tabs	-	Test
NFR-16	Must	Shell and Tabs	-	Test
NFR-17	Must	Shell and Tabs	-	Analysis
NFR-18	Must	Shell and Tabs	-	Inspection
NFR-19	Must	Shell and Tabs	-	Inspection
NFR-20	Should	Shell and Tabs	-	Demo
NFR-21	Must	FormatCodec	ADR-002	Inspection
NFR-22	Must	IPC bridge	ADR-001	Inspection
NFR-23	Should	Build and release	-	Analysis
NFR-24	Must	Build and release	ADR-008	Test
NFR-25	Must	Build and release	-	Test
NFR-26	Must	PreferencesStore	-	Inspection
NFR-27	Must	Build and release	ADR-008	Test
NFR-28	Must	Build and release	-	Inspection
NFR-29	Must	Build and release	ADR-005	Inspection
NFR-30	Must	Build and release	-	Inspection
NFR-31	Must	Build and release	ADR-008	Demo

10.1 Quality scenarios

QS	Component	View	Verification
QS-01	EditorCore, FormulaRenderer	Figure 2	Test: input-latency trace
QS-02	FileService	Figures 2, 3	Test: kill during save
QS-03	FormatCodec, ClipboardBridge	Figure 5	Test: hostile sample set (section 6.10)

QS	Component	View	Verification
			under Node and in the app
QS-04	AutosaveService, FileService	Figures 3, 8	Test: change the file in Notepad
QS-05	ExportService	Figure 4	Test: open in Word
QS-06	SignatureStore	Figure 2	Inspection: read the file as another user
QS-07	IPC bridge, FileService	Figure 2	Test: fake renderer under Node (H-30)

11. Open decisions

These SRS open decisions affect the architecture. IDs match the SRS.

ID	Question	Proposed default	Status
OD-02	File extension and format version	.obk, format 1, after checking it is not widely used	Open
OD-03	Autosave interval	Within 30 s of the last edit, as a target	Open
OD-05	Diagram engine and syntax	Mermaid, bundled, offline; license to verify	Open
OD-08	Exact bundled fonts and licenses	LibreOffice Latin set plus Dancing Script, each license checked	Open
OD-11	Supported Windows versions	Windows 11 and Windows 10 22H2, x64	Open
OD-07	Crash safety for never-saved tabs	Should: recovery copy every 30 s	Decided 2026-10-23
OD-12	Open from File Explorer	Should: one window, new tab per file	Decided 2026-10-23
OD-13	Do table references follow row and column insert and delete?	Rewrite them; a reference to a deleted cell becomes #REF!	Decided 2026-10-23
OD-14	How are other format versions opened?	Convert older versions on open, refuse newer ones, and confirm before the first save upgrades a file	Decided 2026-10-23
OD-15	How are saved signatures stored?	A PNG for every kind, one encrypted file each	Decided 2026-10-23
OD-16	A recovery copy neither restored nor discarded	Keep it and offer it at every launch	Decided 2026-10-23
OD-17	Restore tabs on launch	Keep; open paths are stored in session.json (section 9)	Decided 2026-10-23
OD-18	Animated GIF and WebP images	Keep the original and play it in the editor; PDF, Word, and print use the first frame; Markdown keeps the original file	Decided 2026-10-23
OD-19	Mermaid security level	strict, which a diagram cannot change	Decided 2026-10-23
OD-20	Open link	Keep; http, https, and mailto only, after a prompt	Decided 2026-10-23
OD-21	Uncompressed size caps for .obk files	document.json 25 MB, each image 20	Decided 2026-10-

ID	Question	Proposed default	Status
		MB, 250 MB in total	23

12. Maintenance and references

Item	Detail
Author	Filliciano Plong
Version and date	1.1, October 23, 2026
Based on	SRS version 1.1
Update when	An ADR changes status, a component is added or renamed, or a spike changes a decision
Security updates	Lockfile and dependency audit in every build; fuses checked in the build; Electron security releases shipped within 30 days (target)
Diagram sources	docs/diagrams/*.puml, rendered to docs/images/; shipped in Plong_Obelisk_DiagramSources.zip

12.1 References

- Obelisk SRS (Plong_Obelisk_SRS.docx), version 1.1.
- arc42 template: arc42.org. C4 model: c4model.com.
- Electron security checklist: electronjs.org/docs/latest/tutorial/security.
- Electron safeStorage: electronjs.org/docs/latest/api/safe-storage.
- Electron app.requestSingleInstanceLock: electronjs.org/docs/latest/api/app.
- MSIX file system behavior: learn.microsoft.com/windows/msix/desktop/desktop-to-uwp-behind-the-scenes.
- mathml2omml: npmjs.com/package/mathml2omml.
- KaTeX options: katex.org/docs/options.
- Mermaid configuration schema: mermaid.js.org/config/schema-docs/config.